What's New in RDF 1.2

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Abstract

This document provides the reader with a summary of changes to RDF introduced in RDF version 1.2.

Status of This Document

This section describes the status of this document at the time of its publication. A list of current W3C publications and the latest revision of this technical report can be found in the W3C standards and drafts index.

This document is intended to provide the reader with a summary of changes to RDF introduced in RDF version 1.2.

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§ Set of Documents

This document is one of ten RDF 1.2 and twelve SPARQL 1.2 documents produced by the RDF-star Working Group.

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§ 1. Introduction

This document is informative in nature. Its purpose is to provide a summary of differences between RDF versions 1.1 and 1.2 and to introduce new additions in a very brief manner.

Readers new to RDF should start with the *RDF 1.2 Primer* [*RDF12-PRIMER*] and then move on to the specifications in which they are most interested. This document is meant to serve as a guide for those already familiar with RDF 1.1 who wish to understand changes in version 1.2.

Normative specifications of RDF can be found in the following documents:

- A document describing the basic concepts underlying RDF, as well as abstract syntax (*RDF 1.2 Concepts and Abstract Data Model*) [*RDF12-CONCEPTS*]
- A document describing the formal model-theoretic semantics of RDF (*RDF 1.2 Semantics*) [*RDF12-SEMANTICS*]
- Specifications of concrete syntaxes for RDF, such as Turtle [*RDF12-TURTLE*], TriG [*RDF12-TRIG*], N-Triples [*RDF12-N-TRIPLES*], N-Quads [*RDF12-N-QUADS*] and JSON-LD [*JSON-LD11*]. RDFa [*RDFA-PRIMER*] is also a concrete syntax for RDF, but it was not defined by the RDF Working Group. The 2004 RDF/XML syntax was updated to be in line with the RDF 1.2 specifications [*RDF12-XML*].

- A document describing RDF Schema [*RDF12-SCHEMA*], which provides a data-modeling vocabulary for RDF data.

The following prefixes are used in this document:

Prefixes and IRIs

Namespace prefix	**Namespace IRI**	**RDF vocabulary**
rdf	`http://www.w3.org/1999/02/22-rdf-syntax-ns#`	The RDF built-in vocabulary [*RDF12-SCHEMA*]
xsd	`http://www.w3.org/2001/XMLSchema#`	The RDF-compatible XML Schema datatypes

§ 2. Abstract Syntax

§ 3. New Serialization Formats

RDF 1.1 introduces a number of new serialization formats. RDF 1.1 Concepts and Abstract Syntax makes it clear that RDF/XML is no longer the only recommended serialization format; RDF itself should be considered to be the data model (the abstract syntax), not any particular serialization.

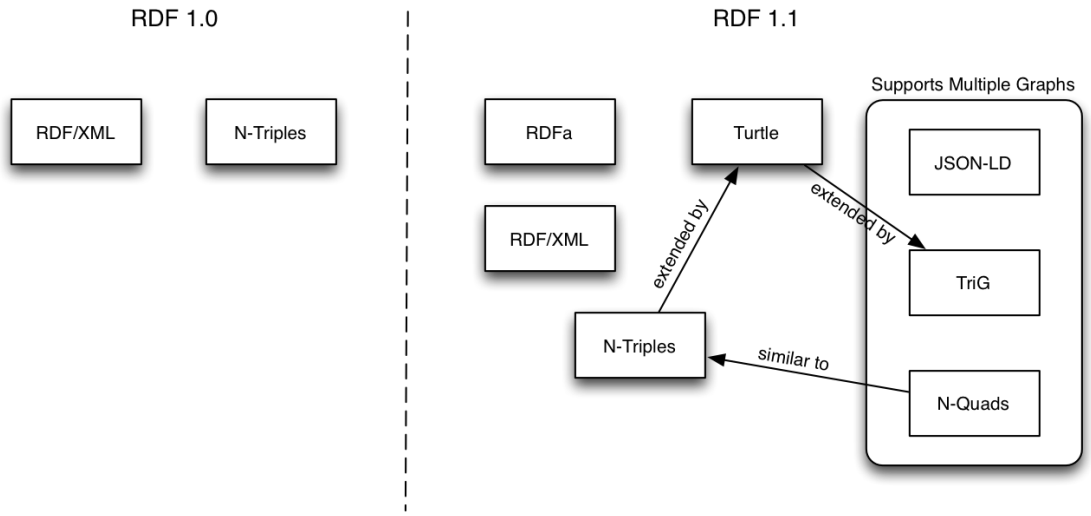


Figure 1 RDF 1.0 and 1.1 serialization formats

§ 4. Semantics

Most of the changes between RDF and RDF 1.1 do not have any effect on implementations of entailment.

Datatype entailment formally refers to a set of 'recognized' datatypes, replacing datatype maps in RDF 1.0 Semantics, but this does not have any effect on implementation.

Datatype entailment formally refers to a set of 'recognized' datatype IRIs. The RDF 1.0 Semantics used the concept of a datatype map: in the new semantic description, this is the mapping from recognized IRIs to the datatypes they identify. This change does not have any effect on implementation or semantic entailments.

RDF entailment has two required datatypes `xsd:string` and `rdf:langString` which must be recognized, but this doesn't appreciably add to RDF entailment as these two datatypes replace plain literals.

One change that does affect entailment is that graphs containing invalid literals (e.g., `"a"^^xsd:integer`) are immediately inconsistent for recognized datatypes, even in sub-RDFS entailment regimes.

RDF 1.1 includes RDF Datasets. However, the semantics of RDF Datasets in RDF 1.1 is minimal and entailment per se is only defined on RDF graphs so there are no changes here.

A. Acknowledgments

A.1 Acknowledgments for RDF 1.1

This section is non-normative.

The editor gratefully acknowledges the members of the RDF Working Group who contributed to this document, including Richard Cyganiak, Gavin Garothers, Pat Hayes, Sandro Hawke, Gregg Kellogg, Markus Lanthaler, Peter Patel-Schneider, Eric Prud-hommeaux, Guus Schreiber and Manu Sporny.

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> EDITOR'S NOTE
>
> Recognize members of the Task Force? Not an easy to find list of contributors.

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